# HW06 - AI-assisted API Testing

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Course: Software Testing

SUT: EShop - <https://github.com/ttbhanh/eshop-sut>

Repository: <https://github.com/KieuDuyennn/software__testing/tree/hw6/HW06>

Execution date: 2026-08-23

## 1. Scope and API selection

The three graded pool selections are API 1 (Pool A), API 3 (Pool B), and API 4 (Pool C). API 2 is additional Pool-A coverage and is reported separately so it does not obscure compliance with the one-per-pool rule.

| API | Pool | Requirement | Endpoint | Role in submission |
|---|---|---|---|---|
| API 1 | A | FR-01 Account registration | `POST /api/register` | Graded Pool A |
| API 2 | A | FR-06 Product detail | `GET /api/products/:id` | Additional coverage |
| API 3 | B | FR-11 Order history | `GET /api/orders/my-orders`, `GET /api/orders/:id`, cancellation | Graded Pool B |
| API 4 | C | FR-13 Dashboard | `GET /api/admin/orders`, admin status transition | Graded Pool C |

FR-13 has no dedicated dashboard endpoint. The frontend derives order count and revenue (sum of `total_amount` for `status = delivered`) from `GET /api/admin/orders`, so this is the tested API.

Group uniqueness: pending the student's dated confirmation with group members. This cannot be inferred from the repository and must be added before submission.

## 2. Test environment and reproducibility

| Item | Value |
|---|---|
| Backend | Node.js + Express + SQLite, freshly reseeded per run |
| Base URL | `http://localhost:3000` |
| Runner | Postman collection format + Newman 6.2.1 |
| Reports | Newman JSON + `newman-reporter-htmlextra` HTML |
| Mandatory identity | `X-Student-Id: 23127184`, injected and asserted at collection level |
| Rate limiting | `LOADTEST=1` during suite runs to avoid invalid HTTP 429 noise |
| Full run log | `evidence/newman-console/suite_full_20260823-223623.log` |
| Gate run log | `evidence/newman-console/suite_gate_20260823-225337.log` |

Reproduce locally:

```
npm install
npm run sut:install
.\scripts\Invoke-ApiTests.ps1
.\scripts\Invoke-ApiTests.ps1 -Mode gate
```

## 3. AI-first method and human-review control

Each API followed four traceable stages:

1. Generate partitions for every parameter, state transitions, SEC-01..SEC-07,

and strict response schemas.

2. Audit every generated case as VALID, INVALID, or INCOMPLETE against the

specification and requirements, not against observed SUT behavior.

3. Add five student-designed cases targeting blind spots found in the audit.

4. Execute on a fresh database, then map failures to root-cause bugs rather

than treating every failed assertion as a separate defect.

One generated case was INVALID: `A2-DP-006` duplicated product-id 5. It was replaced with a percent-encoded-id case. Twenty-two cases were INCOMPLETE because the specification could justify only a weak safety oracle or omitted the behavior entirely. The remaining 363 final cases were VALID. Full reasoning for all 386 rows is in `docs/phases/*/02-audit.md`.

## 4. Results summary

| API | AI-generated | Student-added | Total | Fully passed | Failed cases | Assertions passed | Assertions total |
|---|---|---|---|---|---|---|---|
| API 1 - FR-01 | 121 | 5 | 126 | 69 | 57 | 544 | 606 |
| API 2 - FR-06 | 83 | 5 | 88 | 54 | 34 | 390 | 433 |
| API 3 - FR-11 | 89 | 5 | 94 | 84 | 10 | 407 | 418 |
| API 4 - FR-13 | 73 | 5 | 78 | 68 | 10 | 333 | 345 |
| Total | 366 | 20 | 386 | 275 | 111 | 1,674 | 1,802 |

The 128 failed assertions are expected defect-revealing results. They are clustered into 16 confirmed bugs. The separate regression gate contains cases that the current SUT satisfies and passed 1,262/1,262 assertions.

## 5. API 1 - FR-01 Account registration

Generation produced 121 cases across name/email/password domains, registration state, SEC controls, and response schemas. Audit found ambiguous limits and one misattributed security case: `A1-SEC-013` originally blamed injected role data, but `A1-SEC-012` proves that field is ignored. It was corrected into a direct admin-authorization test.

Student additions cover tab/CRLF email controls, NUL in name, JSON media type with charset, and numeric `confirmPassword`. Three of these produced new evidence for the existing validation root cause. The run confirmed BUG-01, BUG-02, BUG-06, and BUG-07 through BUG-12. Detailed registers:

- `docs/phases/api1-fr01-register/02-audit.md`
- `docs/phases/api1-fr01-register/03-extend.md`
- `docs/phases/api1-fr01-register/04-execute.md`

## 6. API 2 - FR-06 Product detail (additional)

The corrected final suite has 88 cases. Added cases test URL-decoded valid ids, path/query precedence, explicit JSON content negotiation, double encoding, and full-width Unicode digits. The run confirms three root causes: unknown or malformed ids return `200 {}` (BUG-03), even ids expose `price` as a string (BUG-04), and product write routes lack authentication/role checks (BUG-13).

Detailed registers are under `docs/phases/api2-fr06-product-detail/`.

## 7. API 3 - FR-11 Order history

The 94-case suite uses real two-user fixtures and walks the complete FR-10 state machine. Student additions emphasize post-conditions and cross-route consistency. `A3-HR-001` shows that the forbidden shipping cancellation not only returns the wrong status but actually changes the order to `canceled`.

Confirmed root causes: anonymous/cross-user order detail IDOR (BUG-05), missing admin role checks (BUG-06), deleted-account JWTs remain usable (BUG-14), shipping cancellation is accepted (BUG-15), and `canceled -> delivered` is accepted (BUG-16).

Detailed registers are under `docs/phases/api3-fr11-order-history/`.

## 8. API 4 - FR-13 Admin dashboard

The 78-case suite validates authorization, revenue/count projections, every legal and illegal order transition, and the joined response schema. Added cases test aggregation additivity, unique order ids, same-state terminal replay, authorization atomicity, and one-checkout/one-row count integrity.

`A4-HR-004` independently proves a user token can make a legal admin transition and mutate `confirmed -> shipping`. `A4-ST-016` connects the invalid `canceled -> delivered` transition to false delivered revenue. These failures map to BUG-06 and BUG-16.

Detailed registers are under `docs/phases/api4-fr13-admin-orders/`.

## 9. Postman features

Verified locally: collections, nested folders, local/CI/mock environments, environment and global variables, secret variable metadata, collection- and request-level pre-request scripts, `pm.sendRequest` fixture chains, token capture, strict JSON-schema assertions, dynamic values, Newman CLI, HTML/JSON reporting, selective regression gates, and CI configuration.

The signed-in Postman workspace contains all four HW06 collections and the `EShop - Local (23127184)` environment; both views are captured under `evidence/postman-cloud/`. A real Postman Desktop request to `http://localhost:3000/api/products/2` returned HTTP 200, and its Console image shows `[HW06] X-Student-Id=23127184`. A public API2 Mock Server is also running and has recorded a real call. Collection Runner, Monitor and a successful mock example response remain pending.

## 10. CI/CD

The workflow separates a blocking green regression gate from a non-blocking full defect-discovery suite. The local equivalent is verified green at 1,262/1,262. The initial pushed run is authentic red evidence caused by two Linux-unstable stateful cases; the reviewed follow-up gate passed green in GitHub Actions. Both SHAs and run URLs are in `docs/cicd/CI_CD_REPORT.md`.

## 11. Bug summary

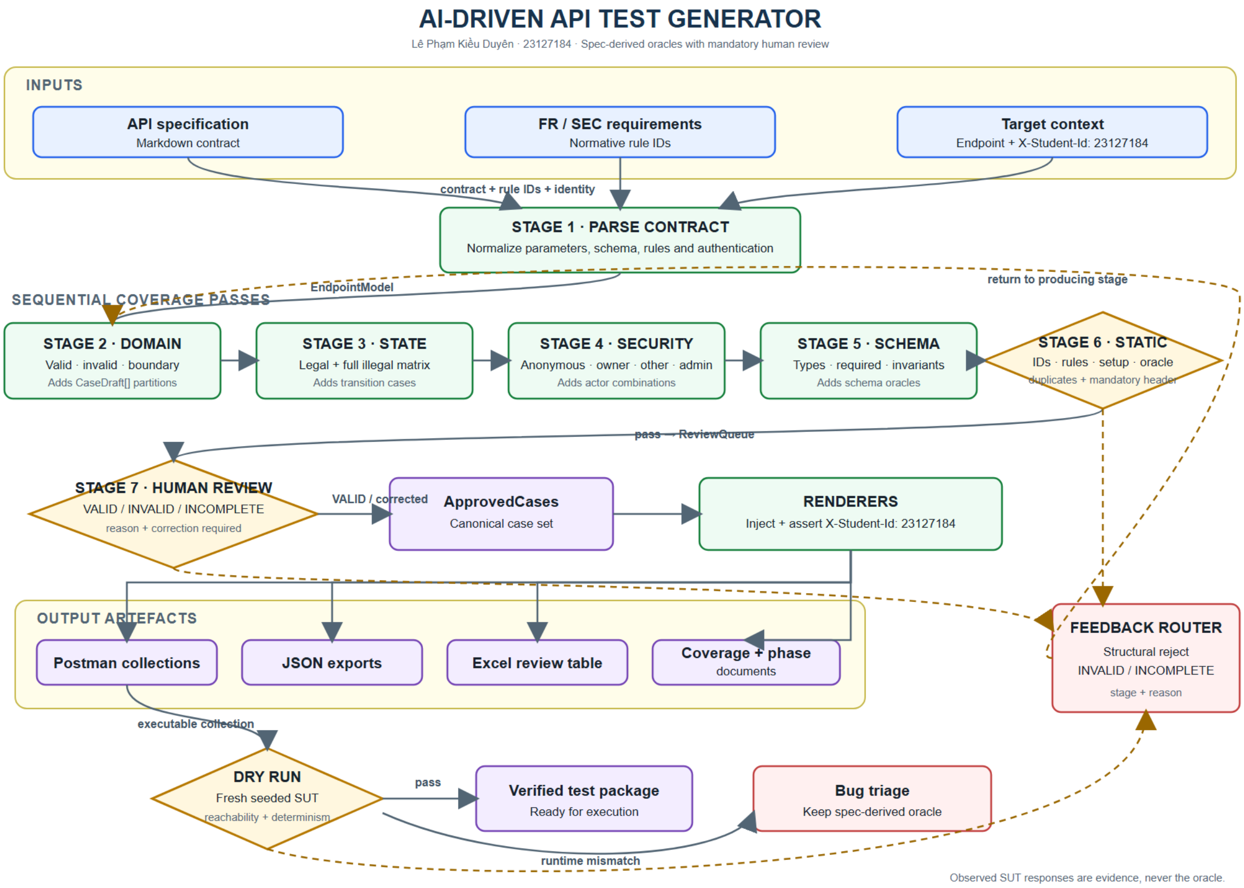
| Range | Count | Main themes |
|---|---|---|
| BUG-01..BUG-12 | 12 | Registration validation, plaintext passwords, product response contract, IDOR, admin authorization, error handling |
| BUG-13..BUG-16 | 4 | Product write authorization, stale JWT, invalid cancellation, terminal-state violation |
| Total | 16 | Full reproduction steps and Newman messages in `docs/bugs/BUG_REPORT.md` |

Real GitHub Issue URLs are recorded in the bug report. Signed-in issue-page screenshots for #66-#70 and real green/red Actions summaries are stored under `evidence/screenshots/`; `evidence/EVIDENCE_INDEX.md` maps every image to its visible verification fields.

## 12. AI-driven test generator (G9.5)

The generator uses a normalized endpoint model, four coverage passes, a human-review queue, executable rendering, a dry-run validator, and a feedback loop. Design and pseudocode are in `docs/design/`. A reusable implementation is under `.claude/skills/api-test-generator/`.

The generator flow is exported at `docs/design/diagram/generator-design.png`, with editable Mermaid source beside it. It shows the three contract inputs, seven design stages, mandatory VALID / INVALID / INCOMPLETE human-review gate, rendered artifacts, dry-run validation, bug triage, and feedback paths. The student must review/adapt the source and be able to explain every decision in the narrated demonstration video.




AI-driven API test generator pipeline

## 13. AI Critique

See `docs/AI_CRITIQUE.md` (258 words). It identifies the duplicated case, misattributed security failure, and missing post-condition oracles, and derives the rule that AI output is a review queue rather than evidence.

## 14. AI Audit and evidence integrity

The mandatory interaction log is in `docs/ai-audit/AI_AUDIT.md`. Local Newman JSON, HTML, and timestamped console logs are real execution artifacts. The following are intentionally pending because they require the student/device: group uniqueness confirmation, Collection Runner, Monitor, a successful mock example response, final student authorship review of the editable diagram, and the narrated YouTube video. The mandatory executed Postman Console evidence is complete.